



Modelling Financial Markets

Seminar 15: LLMs and Sentiment Analysis — Hands-on



Daniel Traian PELE

Drd. Antoaneta Amza, antoaneta.amza@csie.ase.ro

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AUC: Area Under the (ROC) Curve
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAR: Cumulative Abnormal Return
- ▣ CI: Confidence Interval
- ▣ ECE: Expected Calibration Error
- ▣ FNSPID: Financial News and Stock Price Integration Dataset
- ▣ GPT-4: Generative Pre-trained Transformer 4
- ▣ LLM: Large Language Model
- ▣ LM: Loughran-McDonald (financial sentiment dictionary)
- ▣ NLL: Negative Log-Likelihood
- ▣ OLS: Ordinary Least Squares
- ▣ PPI: Prediction-Powered Inference
- ▣ SD: Standard Deviation
- ▣ SE: Standard Error
- ▣ TF-IDF: Term Frequency - Inverse Document Frequency
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)

Today's Question and Required Outputs

- Question of the seminar: **can a language model read financial sentiment, and does that sentiment predict returns one can still trade?**
- Route in class (about 2×50 minutes of work, the rest discussion)
 - ▶ A1, A3, A7 on paper: how a misclassified label attenuates a slope, how precise a paired accuracy test is, at what cost a strategy stops earning
 - ▶ B1: is FinBERT better than a dictionary? B5: do headlines predict returns or describe them? B7: does the LLM remember the market?
 - ▶ C2: find the three errors in an AI answer; B2 at home
- Required outputs (checklist on the last core slide): the A answers, one table and one figure per B exercise, the C2 corrections, one interpretation sentence per exercise
- Notebook for today: [open the seminar notebook in Colab](#); every exercise has a section with the same label
- Seminar: Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

Exercise Map

Exercise	Question	Type	Model
A1	how much does a misclassified dummy shrink a slope?	Solved	–
A3	how precise, and how powerful, is a paired accuracy test?	Solved	–
A7	at what cost per trade does the strategy stop earning?	Solved	–
B1	is FinBERT more accurate than the LM dictionary?	Solved	–
B5	do headlines predict returns, or describe them?	Solved	–
B7	does the LLM remember the market?	Solved	–
C2	what is wrong in an AI answer?	Proposed	B1, B5, B7
B2	is a 7-billion-parameter LLM better than FinBERT? (homework)	Proposed	B1
Extension packs	A2, A4, A5, A6, A8, B3, B4, B6, B8–B11, C1, C3 (optional, after the seminar)	both	named on each task

- ▣ **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- ▣ Part A: derivations on paper; Part B: verification and inference on real data, each ending with an interpretation question; Part C: open questions and AI critique

Data Manifest

Data	Content	Saved scores	Used in
Twitter Financial News (validation set, Hugging Face Hub)	2 388 headlines from 2022: 1 566 neutral, 475 positive, 347 negative	LM, FinBERT, Qwen2.5 (P1–P3), MiniLM	A1–A5, A8, B1–B4, B8, B9, C2
Financial PhraseBank Malo et al. (2014)	4 846 news sentences, 2 264 with full agreement	same scores	B3
FNSPID Dong, Fan & Peng (2024)	106 820 headlines on 16 US stocks, 27.10.2009–18.12.2023	daily LM, FinBERT, Qwen2.5-7B scores	A7, A8, B5, B6, B10, B11, C1–C3
Stocks and SPY	data/market/<SYMBOL>.csv, adjusted_close, 2009–2023	–	B5, B6, C1, C3
S&P 500	GSPC.INDX, close, monthly, Jan 2000 – Aug 2026	Qwen2.5 answers about each month	B7, C2

- The scores are saved tables in [Quantlets/Ch_15](#) (ch15_twitter_scores.csv, ch15_phrasebank_scores.csv, ch15_news_daily.csv, ch15_memory_monthly.csv): no language model has to run in class
- C3 (extension pack) needs a new scoring run of the FNSPID headlines

Setup: Run and Verify

```
# notebook section "Setup" defines the loaders
pb, tw = eval_texts()    # PhraseBank, Twitter
P, rets = news_panel()   # FNSPID stock-days
print(len(pb), len(tw))
print(tw['label'].value_counts())
print(P.shape,
      P.index.get_level_values('ticker').nunique())
```

- Expected output
 - ▶ 4 846 and 2 388 rows
 - ▶ neutral 1 566, positive 475, negative 347
 - ▶ (18 282, 19) and 16 stocks
- Each row of `tw` holds the text, the human label and the probabilities of every model (e.g. `finbert_negative`, `qwen7B_positive`)
- If your counts differ, stop and check the files before any exercise

Notation Used in Every Exercise

- Dictionary tone $\tau = (P - N)/(P + N)$, with P , N the counts of positive and negative words; LM: Loughran–McDonald financial lists [Loughran & McDonald \(2011\)](#)
- Model probabilities $p_k = e^{z_k/T} / \sum_j e^{z_j/T}$ (z_k : logit of answer k , T : temperature); label $\arg \max_k p_k$; net score $s = p_{\text{pos}} - p_{\text{neg}}$; FinBERT [Araci \(2019\)](#), Qwen2.5 [Qwen Team \(2024\)](#)
- Two classifiers A and B on the same texts: n_{01} = A wrong and B right, n_{10} = A right and B wrong; McNemar test [McNemar \(1947\)](#): under H_0 , $n_{01} \sim \text{Bin}(n_{01} + n_{10}, 0.5)$
- Day d : the first trading day on or after the headline date; $d + 2$: the first return surely after publication; h : days after d
- bp: basis point, 0.01%; SE: standard error; CI: 95% confidence interval; t_{NW} : Newey–West t [Newey & West \(1987\)](#); clustered SE: [Petersen \(2009\)](#)

A1 [Solved]: Attenuation from Misclassification — Task

- **Question:** a regression uses the FinBERT label "negative" instead of the truth; by how much is the slope shrunk?
- **Data**
 - ▶ Twitter Financial News, validation set: $N = 2\,388$ headlines from 2022, each labelled negative, neutral or positive; the human label is treated as the truth
 - ▶ Model $r = a + \beta s^* + \varepsilon$, $s^* = 1$ if the headline is truly negative; we observe only $\hat{s} = 1$ when FinBERT predicts negative
 - ▶ Non-differential error: given s^* , $\hat{s}$ says nothing about r , so $\text{Cov}(\varepsilon, \hat{s}) = 0$
 - ▶ $\pi = P(s^* = 1)$, $p = P(\hat{s} = 1)$, false alarms
 $\alpha_0 = P(\hat{s} = 1 \mid s^* = 0)$, misses $\alpha_1 = P(\hat{s} = 0 \mid s^* = 1)$
- **Tasks**
 - (a) Show that $\text{plim } \hat{b} = \beta \text{Cov}(s^*, \hat{s}) / \text{Var}(\hat{s})$.
 - (b) Derive $\lambda = \pi(1 - \pi)(1 - \alpha_0 - \alpha_1) / [p(1 - p)]$.
 - (c) Compute π , p , α_0 , α_1 from the matrix, then λ to three decimals.
 - (d) Compute $\text{Cov}(u, s^*)$ for $u = \hat{s} - s^*$: is the error classical?
- **Report:** λ , $1/\lambda$ and the sign of $\text{Cov}(u, s^*)$
- **Interpretation:** what share of a true effect survives in a slope on the FinBERT dummy?

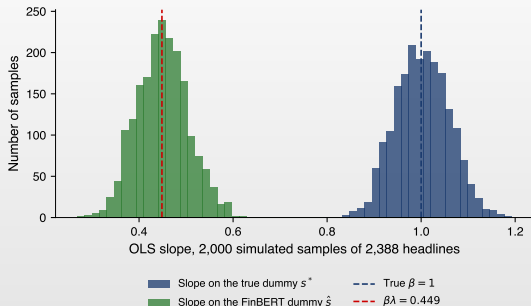
FinBERT confusion matrix, Twitter

True \ predicted	neg	neu	pos	Total
neg	264	71	12	347
neu	213	1 184	169	1 566
pos	58	133	284	475
Total	535	1 388	465	2 388

A1 [Solved]: Solution (1/2)

- (a) For one regressor $\text{plim } \hat{b} = \text{Cov}(r, \hat{s}) / \text{Var}(\hat{s})$ and $\text{Cov}(r, \hat{s}) = \beta \text{Cov}(s^*, \hat{s}) + \text{Cov}(\varepsilon, \hat{s})$; the last term is 0 by the non-differential assumption
- (b) Both variables are binary: $\text{Cov}(s^*, \hat{s}) = P(s^* = 1, \hat{s} = 1) - \pi p = \pi(1 - \alpha_1) - \pi p$
- Total probability: $p = \pi(1 - \alpha_1) + (1 - \pi)\alpha_0$; substituting, $\text{Cov}(s^*, \hat{s}) = \pi(1 - \pi)(1 - \alpha_0 - \alpha_1)$
- $\hat{s}$ is Bernoulli, so $\text{Var}(\hat{s}) = p(1 - p)$; the ratio of the two is λ

A1 [Solved]: Solution (2/2)

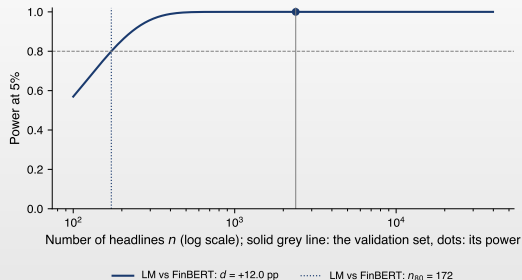


- (c) $\pi = 347/2\,388 = 0.1453$, $p = 535/2\,388 = 0.2240$, $\alpha_1 = 1 - 264/347 = 0.2392$, $\alpha_0 = 271/2\,041 = 0.1328$
- $\lambda = \frac{0.1242 \times 0.6280}{0.2240 \times (1 - 0.2240)} = 0.449$, so $\beta = 2.23 \hat{\beta}$
- (d) $\text{Cov}(u, s^*) = 0.0780 - 0.1242 = -0.0462 < 0$: the error is negatively correlated with the truth, not classical [Aigner \(1973\)](#)
- Chart: 2 000 simulated samples with these rates and $\beta = 1$; the slope on $\hat{s}$ averages 0.450 (SD 0.054), as λ predicts
- Interpretation: a slope on the FinBERT dummy keeps only about 45% of the true effect

A3 [Solved]: Paired Accuracy Difference: Variance, Exact Test, Power — Task

- ▣ **Question:** how precise is the accuracy gap between two classifiers scored on the same headlines, and how many headlines does it need?
- ▣ **Inputs**
 - ▶ Twitter validation set, $n = 2\,388$ independent headlines; $A = \text{LM dictionary}$, $B = \text{FinBERT}$
 - ▶ Discordant counts: $n_{01} = 539$ (A wrong, B right), $n_{10} = 252$ (A right, B wrong)
 - ▶ $D_i = 1$ if A wrong and B right, -1 if A right and B wrong, 0 otherwise; $d = ED_i = \text{acc}_B - \text{acc}_A$, $q = P(D_i \neq 0)$
- ▣ **Tasks**
 - (a) Show $\hat{d} = (n_{01} - n_{10})/n$ and derive $\text{Var}(\hat{d}) = (q - d^2)/n$.
 - (b) Compute $\hat{d}$, SE, the 95% CI and $z = \hat{d}/\text{SE}$.
 - (c) Run the exact McNemar test at 5% and state the decision.
 - (d) Compute the power $\Phi(|d|\sqrt{n}/\sqrt{q - d^2} - 1.96)$ if the true d is $\hat{d}$, and $n_{80} = (1.96 + 0.84)^2(q - d^2)/d^2$.
- ▣ **Report:** $\hat{d}$, SE, CI, z , exact p , the decision, the power and n_{80}
- ▣ **Interpretation:** which headlines determine the variance of $\hat{d}$?

A3 [Solved]: Solution

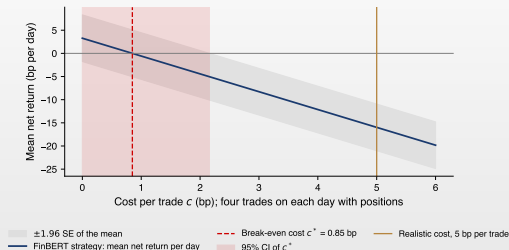


- (a) $\hat{d} = \frac{1}{n} \sum_i D_i = (n_{01} - n_{10})/n$; $D_i^2 = 1$ exactly when one is right and the other wrong, so $ED_i^2 = q$, $\text{Var}(D_i) = q - d^2$ and $\text{Var}(\hat{d}) = (q - d^2)/n$
- (b) $\hat{d} = (539 - 252)/2388 = 0.1202$,
 $\hat{q} = 791/2388 = 0.3312$, $\text{SE} = 1.15$ pp, $\text{CI} [9.76; 14.28]$ pp, $z = 10.43$
- (c) Under H_0 we expect 395.5 on each side; exact $p < 0.001$: reject equal accuracy
- (d) Power $\Phi(+8.47) \approx 100\%$;
 $n_{80} = 7.84 \times (0.3312 - 0.01444)/0.01444 \approx 170$ headlines (chart)
- Interpretation: only the discordant headlines (q) enter the variance; a 12-point gap is detectable with a few hundred headlines

A7 [Solved]: Break-Even Cost with a Standard Error — Task

- **Question:** at what cost per trade does the FinBERT strategy stop earning, and how precisely do we know that cost?
- **Inputs**
 - ▶ Strategy of Lecture 15: at the close of $d + 1$ take $\text{sign}(s_{i,d})$ in every stock with news on day d , hedged with SPY, closed one day later; four trades on each day with positions; 3 568 trading days
 - ▶ FinBERT signal: mean gross return over all days $\hat{\mu} = 3.27$ bp, $t_{NW} = 1.27$ ($L = 8$ lags), share of days with positions $a = 0.962$, gross Sharpe ratio $\sqrt{252} \hat{\mu} / \sigma = 0.36$
- **Tasks**
 - (a) Write the mean net daily return and the net Sharpe ratio as functions of the cost c per trade.
 - (b) Derive the break-even cost c^* at which the mean net return is zero.
 - (c) Compute $\text{SE}(\hat{\mu}) = \hat{\mu} / t$, then $\text{SE}(c^*)$ by the delta method and the 95% CI of c^* .
 - (d) Recover σ from the gross Sharpe ratio and compute the net Sharpe ratio at $c = 5$ bp.
- **Report:** c^* , its SE and CI, the net Sharpe ratio
- **Interpretation:** is a realistic cost of 5 bp per trade inside the CI of c^* ?

A7 [Solved]: Solution



- (a) On a share a of days we pay $4c$: $\mu_{\text{net}} = \mu - 4ac$,
 $SR_{\text{net}} = \sqrt{252} (\mu - 4ac) / \sigma$, with σ barely changed by costs
- (b) $c^* = \mu / (4a) = 3.27 / 3.85 = 0.85$ bp per trade
- (c) $SE(\hat{\mu}) = 3.27 / 1.27 = 2.57$ bp; with a fixed by the news calendar, $SE(c^*) = 2.57 / 3.85 = 0.67$ bp, CI $[-0.46; 2.16]$ bp
- (d) $\sigma = \sqrt{252} \times 3.27 / 0.36 = 143$ bp;
 $SR_{\text{net}} = \sqrt{252} (3.27 - 3.85 \times 5) / 143 = -1.77$
- Interpretation: 5 bp lies far above the CI of c^* ; with daily turnover the strategy cannot survive realistic costs

 MFM_ch15_sem_partA

B1 [Solved]: Is FinBERT Better than LM? — Task

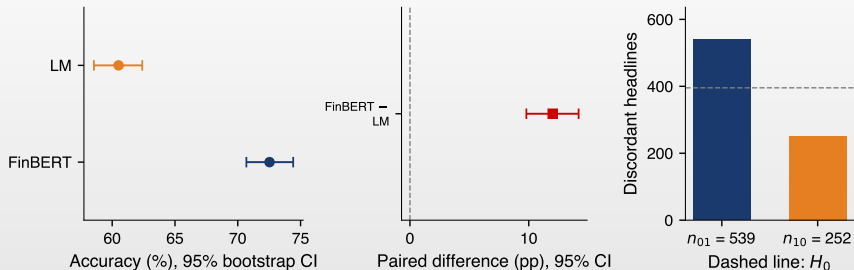
- **Question:** is the fine-tuned FinBERT more accurate than the Loughran–McDonald dictionary on headlines it was not trained on?
- **Data**
 - ▶ Twitter Financial News, validation set: $N = 2\,388$ headlines from 2022, each labelled negative, neutral or positive; the human label is treated as the truth
 - ▶ A = LM dictionary (label = sign of the tone), B = FinBERT (label = largest probability); notebook function `b_pairs`
- **Tasks**
 - (a) Compute each accuracy with a percentile bootstrap 95% CI: resample the headlines 2 000 times (seed 0).
 - (b) Compute the difference $B - A$ with a paired bootstrap CI: the same resampled headlines for both classifiers.
 - (c) Count n_{01} , n_{10} ; run the exact McNemar test and $(n_{01} - n_{10})^2 / (n_{01} + n_{10})$ against χ^2_1 (3.84).
- **Report:** a table of the two accuracies, their CIs and the difference; n_{01} , n_{10} , both p -values and the decision
- **Interpretation:** why is the paired test more powerful than comparing the two intervals, and does the gap matter for trading?

B1 [Solved]: Solution (1/2)

	Accuracy (%)	CI 95%
LM	60.5	[58.5; 62.4]
FinBERT	72.5	[70.7; 74.4]
Difference (pp)	+12.0	[+9.8; +14.2]

- Step 1: labels of both classifiers on the 2 388 headlines; step 2: 2 000 bootstrap resamples of the headline indices; step 3: both accuracies and their difference in each resample
- (a)–(b) The two intervals do not overlap and the paired CI of the difference excludes 0; it agrees with the analytic CI of A3

B1 [Solved]: Solution (2/2)

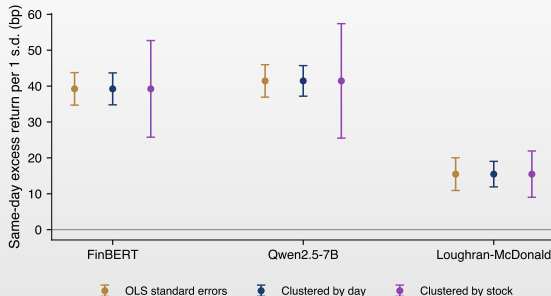


- (c) $n_{01} = 539$, $n_{10} = 252$, 395.5 expected under H_0 ; exact $p < 0.001$; $\chi^2 = 104.1 > 3.84$: reject equal accuracy
- Pairing removes the variation in headline difficulty: headlines both get right, or both get wrong, carry no information
- Interpretation: FinBERT is clearly better than the dictionary outside its training data; a 12-point gain in labelling says nothing yet about returns (B5)

B5 [Solved]: Reaction or Prediction, and Which Standard Errors? — Task

- **Question:** does the daily headline score explain the return of the news day, or forecast the first return one can still trade, and how wide are the honest intervals?
- **Data**
 - ▶ FNSPID panel: 18 282 stock-days with news, 16 US stocks, 27.10.2009–18.12.2023; $z_{i,d}$: daily net score, standardised over the panel; r^e : excess return over SPY (%), from data/market
 - ▶ Scores: FinBERT, Qwen2.5-7B, LM
- **Tasks**
 - (a) Estimate $r_{i,d+h}^e = a + b z_{i,d} + \varepsilon$ by OLS for the three scores and $h = 0, 1, 2$; report b in bp per one SD of the score.
 - (b) Compute three SEs: OLS, clustered by day ($G = 3\,434$), clustered by stock ($G = 16$), with the small-sample factor $\frac{G}{G-1} \frac{n-1}{n-2}$.
 - (c) Compare each t with 1.96, and the stock-clustered t with the t_{15} critical value 2.13.
- **Report:** a chart of the $h = 0$ slopes with the three intervals, and a table of the regressions
- **Interpretation:** why do $h = 0$ and $h = 2$ differ, and can a clustered SE with 16 clusters be trusted?

B5 [Solved]: Solution (1/2)



- Same-day slopes with the 95% intervals of the three standard errors: the stock-clustered interval is the widest
- Clustering by day is close to OLS: with only 5.3 stocks per day there is little common noise
- Clustering by stock multiplies the FinBERT SE at $h = 0$ by 3.0: the slope differs across stocks and each stock's errors are correlated over time

B5 [Solved]: Solution (2/2)

Score	h	b (bp)	SE OLS	SE day	SE stock	t day	t stock
FinBERT	0	+39.2	2.3	2.3	6.9	+17.27	+5.71
FinBERT	2	-0.5	2.2	2.2	1.5	-0.25	-0.35
Qwen2.5-7B	0	+41.4	2.3	2.2	8.1	+19.07	+5.10
Qwen2.5-7B	2	+2.3	2.2	1.9	1.0	+1.19	+2.41
LM	0	+15.5	2.3	1.8	3.3	+8.53	+4.70
LM	2	-0.0	2.2	1.7	0.9	-0.01	-0.03

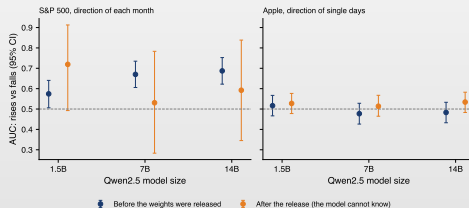
- $h = 0$: large under every SE (t by stock above 4.7): headlines describe the day's move, often the move itself ("shares jump")
- $h = 2$: FinBERT -0.5 bp ($t = -0.25$ by day); Qwen $t = +2.41$ by stock, but $p = 0.061$ by wild cluster bootstrap [Cameron, Gelbach & Miller \(2008\)](#) and 0.37 after Holm over nine regressions (extension B10)
- Interpretation: sentiment explains the news day, not the tradable days; with 16 stocks, inference needs few-cluster methods

B7 [Solved]: Does the Model Remember the Market? — Task

- **Question:** asked without any data, can Qwen2.5 tell the months in which the S&P 500 rose from those in which it fell?
- **Data**
 - ▶ S&P 500 monthly closes (GSPC.INDX), January 2000 to August 2026; a month is "up" if its return is positive
 - ▶ Qwen2.5 1.5B, 7B, 14B (weights published 19 September 2024) asked "Did the S&P 500 index rise or fall during ⟨month year⟩?"; score $P(\text{rise})$
 - ▶ Before: 296 months to August 2024 (185 up); after: 23 months from October 2024 (13 up)
- **Tasks**
 - (a) Compute the $\text{AUC} = U / (n_{\text{up}} n_{\text{down}})$ (U : Mann–Whitney) per model and period, with a 2000-draw bootstrap CI over months.
 - (b) Permutation test for 14B before the release: shuffle the up/down labels 2000 times.
 - (c) Test $\text{AUC}_{\text{before}} = \text{AUC}_{\text{after}}$ with DeLong variances [DeLong et al. \(1988\)](#) and find the smallest drop detectable with 80% power.
- **Report:** the AUC table with CIs, the permutation p , the DeLong z and p , the detectable drop
- **Interpretation:** what does an AUC above 0.5 mean when the model sees no data?

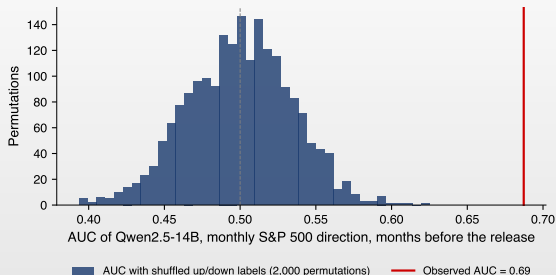
B7 [Solved]: Solution (1/2)

Qwen2.5	AUC before	CI 95%	AUC after	CI 95%
1.5B	0.57	[0.51; 0.64]	0.72	[0.49; 0.91]
7B	0.67	[0.61; 0.73]	0.53	[0.28; 0.78]
14B	0.69	[0.62; 0.75]	0.59	[0.35; 0.84]



- (a) 7B and 14B rank up months above down months far above chance before the release; 1.5B barely; after the release all intervals contain 0.5

B7 [Solved]: Solution (2/2)



- (b) Permutation p for 14B before the release < 0.001 : no shuffled AUC reaches the observed one (chart)
- (c) DeLong: SE before 0.032, after 0.126; $z = 0.73$, $p = 0.47$: do not reject equality; detectable drop
 $2.80 \sqrt{SE_{\text{before}}^2 + SE_0^2} = 0.36$, about 101 post-release months needed
- Interpretation: without data, the discrimination comes from the training text; a backtest on pre-release news mixes forecasting with remembering [Glasserman & Lin \(2023\)](#)

C2 [Proposed]: Critique an AI Answer — the Answer

- **Student prompt:** “I have FNSPID headlines for 16 US stocks, 2009–2023, and daily excess returns vs SPY. Check that Qwen2.5-7B reads sentiment, then test whether its headline sentiment forecasts returns out of sample.”
- **Code of the AI assistant** (qwen_label, qwen_score: the zero-shot scorer of Lecture 15)

```
import pandas as pd, statsmodels.formula.api as smf
from sklearn.metrics import accuracy_score
tw['pred'] = [qwen_label(t) for t in tw['text']] # Twitter validation set
acc = accuracy_score(tw['label'], tw['pred'])
news['day'] = first_trading_day_on_or_after(news['date'])
news['s'] = [qwen_score(t) for t in news['title']] # P(pos) - P(neg)
d = news.groupby(['ticker', 'day'])['s'].mean().reset_index()
d['z'] = (d['s'] - d['s'].mean()) / d['s'].std()
d = d.merge(rets, on=['ticker', 'day']) # rets: ticker, day, xret (%)
fit = smf.ols('xret ~ z', data=d).fit(cov_type='cluster',
                                     cov_kws={'groups': pd.factorize(d['day'])[0]})
```

C2 [Proposed]: Critique an AI Answer — Task

- ▣ **Model:** B1, B5 and B7 [Solved]
- ▣ **Conclusion of the AI assistant**
 - ▶ “Accuracy 74.2%, more than twice the 33% of a random guess among three classes: Qwen2.5-7B reads financial sentiment very well.”
 - ▶ “Slope +41.4 bp per standard deviation of the score, $t = +19.07$ (clustered by day): the day’s headlines forecast the stock’s excess return.”
 - ▶ “Qwen2.5 was never trained on these labels or return series, so the 2009–2023 backtest is fully out of sample.”
- ▣ **Question:** the code runs and the numbers are right, yet the answer contains **three** errors typical of sentiment backtests; which?
- ▣ **Tasks**
 - ▶ (a) Find each error: name the line of code or the sentence and explain why it is wrong.
 - ▶ (b) Say how you would detect it without knowing the answer: a class count, a timestamp check, a release date, a memory test.
 - ▶ (c) Write the corrected code or statement, with the numbers it produces.
 - ▶ (d) Log each error as in [AI_ERRORS.md](#): what the AI answered, what was wrong, how you found out, the correct version.

B2 [Proposed]: Is a 7-Billion-Parameter LLM Better than FinBERT? — Task

- **Model:** B1 [Solved]
- **Question:** does a general open LLM, used zero-shot, beat the fine-tuned FinBERT on headlines neither was trained on?
- **Data**
 - ▶ Twitter Financial News, validation set: $N = 2\,388$ headlines from 2022, each labelled negative, neutral or positive; the human label is treated as the truth
 - ▶ $A = \text{FinBERT}$, $B = \text{Qwen2.5-7B}$ zero-shot with prompt P1; neither used any Twitter label
- **Tasks**
 - (a) Compute each accuracy with a percentile bootstrap 95% CI: resample the headlines 2000 times (seed 0).
 - (b) Compute the difference $B - A$ with a paired bootstrap CI: the same resampled headlines for both classifiers.
 - (c) Count n_{01} , n_{10} ; run the exact McNemar test and $(n_{01} - n_{10})^2 / (n_{01} + n_{10})$ against χ^2_1 (3.84).
- **Report:** the same table, tests and chart as in B1
- **Interpretation:** do the two intervals overlap, does that settle the question, and what would a much larger, slower model still bring?

Submission Checklist and Team Repository

Hand in before the next seminar

- ▣ A1, A3, A7: the derivations and the numbers
- ▣ B1, B5, B7: one table and one figure each, as in the solutions
- ▣ C2: the three errors, how you detected them, the corrected code and your [AI_ERRORS.md](#) entries
- ▣ B2: the table, the tests and the pairs chart
- ▣ One interpretation sentence per exercise

Team repository

- ▣ Choose a text source not covered in class (BVB announcements, Romanian financial news, central-bank statements, crypto posts)
- ▣ Label at least 200 texts with two annotators and report the agreement; evaluate only on a held-out part
- ▣ Log AI use in [AI_USE.md](#) and caught errors in [AI_ERRORS.md](#); extension packs are optional, C1 and C3 are project starts

Extension Packs: What Each One Needs

Pack	Exercises	Prerequisites
1. Measurement error	A2, A6	A1
2. Comparing classifiers	A4, B3 [Solved], B4, B8	A3, B1
3. LLM probabilities	A5 [Solved]	Notation slide, lecture calibration chart
4. Valid inference	B9 [Solved], B10, B11	B5, B7
5. Signals and costs	A8, B6, C1	A7, B5
6. Replication lab	C3	C1, the lecture case study

- Each exercise has its own notebook section; do them in order within a pack
- Useful for the team project: C1 and C3 are possible project starts

A2 [Proposed]: Attenuation for the LM Dictionary — Task

- **Model:** A1 [Solved]
- **Question:** is the weaker same-day LM slope of Lecture 15 (15.5 bp against 39.2 bp for FinBERT) a weaker market reaction, or a worse measurement?
- **Data**
 - ▶ Twitter Financial News, validation set: $N = 2\,388$ headlines from 2022, each labelled negative, neutral or positive; the human label is treated as the truth
 - ▶ Same model and assumption as A1; now $\hat{s} = 1$ when the LM tone of the headline is negative
- **Tasks**
 - (a) Compute π , p , α_0 and α_1 from the LM matrix.
 - (b) Compute λ and $1/\lambda$ with the formula of A1.
 - (c) Compute $\text{Cov}(u, s^*)$: is the LM error classical?
 - (d) Compare λ with FinBERT (A1): which attenuates more, and by what ratio?
- **Report:** the four rates, λ , $1/\lambda$, the ratio

LM confusion matrix, Twitter

True \ predicted	neg	neu	pos	Total
neg	172	166	9	347
neu	247	1 192	127	1 566
pos	45	349	81	475
Total	464	1 707	217	2 388

A6 [Proposed]: Fleiss' κ and One Dawid–Skene Step — Task

- ▣ **Model:** A1 [Solved]
- ▣ **Question:** how much do raters agree beyond chance, and what is the most likely true label of a disputed sentence?
- ▣ **Inputs**
 - ▶ Illustrative votes (made up for the exercise): 6 sentences, $m = 5$ raters each
 - ▶ $P_i = \sum_j n_{ij}(n_{ij} - 1)/[m(m - 1)]$, $\bar{P} = \frac{1}{N} \sum_i P_i$, p_j = share of votes for j ,
 $\bar{P}_e = \sum_j p_j^2$, $\kappa = (\bar{P} - \bar{P}_e)/(1 - \bar{P}_e)$ [Fleiss \(1971\)](#)
- ▣ **Tasks**
 - (a) Compute every P_i and $\bar{P}$.
 - (b) Compute p_j , $\bar{P}_e$ and κ .
 - (c) E-step of [Dawid & Skene \(1979\)](#): each rater gives the true class with probability 0.8 and each other class with 0.1, uniform prior; compute $P(\text{true} = j \mid \text{votes}) \propto \prod_i \pi_{ji}^{n_{ij}}$ for sentences 4 and 5.
 - (d) Compare the posteriors with the vote shares and describe the M-step.
- ▣ **Report:** κ and the two posteriors
- ▣ **Interpretation:** how would you treat the disputed sentences in a test set?

Sentence	neg	neu	pos
1	5	0	0
2	0	5	0
3	0	1	4
4	1	3	1
5	2	3	0
6	0	2	3

A4 [Proposed]: Is "No Difference" a Power Problem? — Task

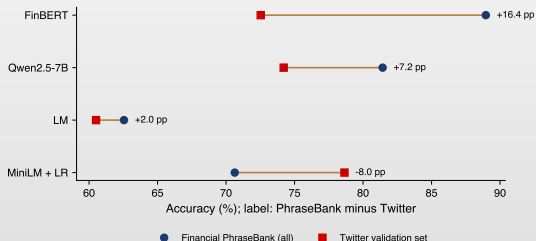
- **Model:** A3 [Solved]
- **Question:** FinBERT and Qwen2.5-7B look tied: is that evidence of equal accuracy, or of a weak test?
- **Inputs**
 - ▶ Twitter validation set, $n = 2\,388$; $A = \text{FinBERT}$, $B = \text{Qwen2.5-7B (prompt P1)}$; $n_{01} = 395$, $n_{10} = 355$
- **Tasks**
 - (a) Compute $\hat{d}$, $\hat{q}$, SE, the 95% CI and z with the formulas of A3.
 - (b) Run the exact McNemar test at 5%.
 - (c) Compute the power at $n = 2\,388$ if the true d is $\hat{d}$, and n_{80} .
 - (d) Use the CI to state which differences the data rule out.
- **Report:** the numbers of (a)–(d)
- **Interpretation:** may you conclude that the two models are equally accurate?

B3 [Solved]: Training-Data Leakage — Task

- ▣ **Question:** how much of FinBERT's accuracy on Financial PhraseBank comes from having been trained on it?
- ▣ **Data**
 - ▶ Financial PhraseBank [Malo et al. \(2014\)](#): 4 846 sentences (2 264 with full agreement); FinBERT was fine-tuned on a split of them (3 101 of 4 845 in [Araci \(2019\)](#))
 - ▶ Twitter validation set, 2 388 headlines unseen by FinBERT; classifiers: FinBERT, Qwen2.5-7B, LM, MiniLM + logistic regression trained on Twitter [Wang et al. \(2020\)](#)
- ▣ **Tasks**
 - (a) Compute accuracy (2 000-draw bootstrap CI) and macro-F1 on PhraseBank with full agreement, all of PhraseBank and Twitter.
 - (b) Compute the drop from all of PhraseBank to Twitter for each classifier.
 - (c) Compute the difference in differences FinBERT minus Qwen2.5-7B with a bootstrap CI (each data set resampled independently).
- ▣ **Report:** a table of accuracies and drops, the difference in differences with its CI
- ▣ **Interpretation:** how do you separate "harder data" from "memory of the training set"?

B3 [Solved]: Solution

Accuracy (%)	PhraseBank, 100% agr.	PhraseBank, all	Twitter	Drop (pp)	Macro-F1, Twitter
FinBERT	97.2	89.0	72.5	16.4	66.8
Qwen2.5-7B	93.6	81.4	74.2	7.2	72.6
LM	65.0	62.5	60.5	2.0	46.2
MiniLM + log. reg.	81.4	70.6	78.6	-8.0	69.9



- (c) $16.4 - 7.2 = 9.2$ pp from unrounded accuracies, bootstrap CI [6.7; 11.9]: excludes 0
- Qwen2.5-7B was not tuned on PhraseBank labels, so its drop measures the harder data; if both would lose the same without overlap (an untested assumption), FinBERT's extra 9.2 points come from overlap
- Interpretation: MiniLM moves the other way because it is trained on Twitter: every model is best at home

B4 [Proposed]: How Many Labels Do You Need? — Task

- **Model:** B3 [Solved]
- **Question:** how many labelled headlines does a simple supervised model need to beat FinBERT?
- **Data**
 - ▶ Twitter Financial News: training set of 9 543 headlines, validation set of 2 388; benchmark FinBERT, 72.5%, no Twitter label used
 - ▶ Logistic regression ($C = 4$) on 384-dimensional MiniLM embeddings; logistic regression ($C = 8$) on TF-IDF features of words and word pairs
- **Tasks**
 - (a) For $n \in \{100, 200, 400, \dots, 6\,400, 9\,543\}$ draw 10 random training subsets, fit both models and average the validation accuracy.
 - (b) Find the smallest n at which each model beats FinBERT on average.
 - (c) Report the SD of the embedding accuracy over the 10 repetitions at $n = 100$.
- **Report:** the learning curve and its table
- **Interpretation:** why do embeddings start higher but grow more slowly?

B8 [Proposed]: Prompt Sensitivity — Task

- ▣ **Model:** B1 [Solved]
- ▣ **Question:** does the answer of the same LLM change when the same question is worded differently?
- ▣ **Data**
 - ▶ Twitter Financial News, validation set: $N = 2\,388$ headlines from 2022, each labelled negative, neutral or positive; the human label is treated as the truth
 - ▶ Qwen2.5-7B and 14B with three prompts: P1 "classify the sentiment . . . for investors", P2 "good, bad or neutral for the stock price", P3 "what is the sentiment of this text"
- ▣ **Tasks**
 - (a) Compute accuracy with a 2 000-draw bootstrap CI and the share of "neutral" answers for each prompt and model.
 - (b) Run the exact McNemar test of P1 against P3 for each model.
 - (c) Compute the share of headlines with the same label under all three prompts.
- ▣ **Report:** a table and a chart of the three prompts
- ▣ **Interpretation:** if you picked the best prompt on this set, what would you be doing?

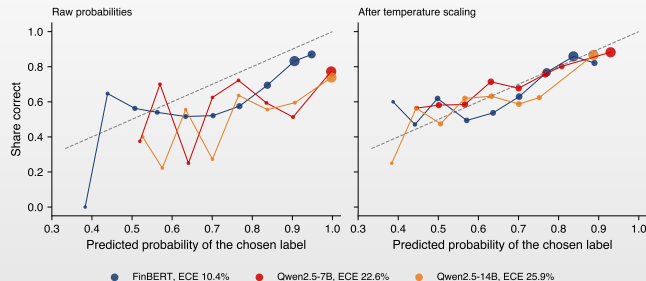
A5 [Solved]: From Label Probabilities to a Score, and the Optimal Temperature — Task

- **Question:** how does the temperature T change an LLM's probabilities, its score and its calibration, and which T fits the data?
- **Inputs**
 - ▶ One Twitter headline labelled **neutral**; Qwen2.5-7B (P1) gives $p_{\text{neg}} = 0.608$, $p_{\text{neu}} = 0.392$, $p_{\text{pos}} = 0.000$ (exactly 1.0×10^{-12})
 - ▶ A random half of the validation set ($n = 1\,194$, seed 15): logits $z_{ik} = \ln p_{ik}$, rescaled $p_{ik}(T) \propto e^{z_{ik}/T}$, loss $\text{NLL}(T) = -\sum_i \ln p_{iy_i}(T)$ Guo et al. (2017)
- **Tasks**
 - (a) Give the label, the net score s and the logit gap $\ln(p_{\text{neu}}/p_{\text{neg}})$ of the headline, then its probabilities and s at $T = 2$.
 - (b) Explain why labels and accuracy do not depend on T , while the Brier score does.
 - (c) Derive $\text{NLL}'(T)$, evaluate the first-order condition at $T = 1$ and solve for T^* (Brent on $[0.5, 50]$).
 - (d) On the other half, compute ECE and Brier before and after scaling.
- **Report:** the label, s , $s(T)$, T^* , ECE and Brier before and after
- **Interpretation:** is the LLM over- or under-confident, and what happens to a rule with a threshold on s ?

A5 [Solved]: Solution (1/2)

- ▣ (a) Largest probability p_{neg} : label negative, wrong; $s = 0.000 - 0.608 = -0.608$; logit gap $\ln(0.392/0.608) = -0.437$
- ▣ At $T = 2$, $p_k(T) \propto \sqrt{p_k}$: $\sqrt{p} = (0.780, 0.626, 1.0 \times 10^{-6})$, sum 1.406, $p(T) = (0.554, 0.446, 0.000)$, $s(T) = -0.554$: same label, score pulled towards 0
- ▣ (b) $z \mapsto z/T$ is increasing for $T > 0$: the order of the logits, hence $\arg \max$ and accuracy, do not change; the probabilities, and with them Brier and log loss, do

A5 [Solved]: Solution (2/2)

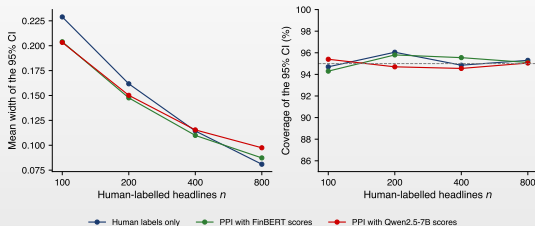


- (c) $\ln p_{iy}(T) = z_{iy}/T - \ln \sum_k e^{z_{ik}/T}$ gives $\text{NLL}'(T) = \frac{1}{T^2} \sum_i (z_{iy_i} - \sum_k p_{ik}(T) z_{ik})$
- At $T = 1$ the average bracket is $-2.47 < 0$, so NLL falls as T grows; the root is $T^* = 8.43$
- (d) ECE falls from 22.6% to 4.3%, Brier from 0.468 to 0.366 (chart: raw and rescaled reliability)
- Interpretation: raw LLM probabilities are heavily overconfident; a threshold on s selects different headlines after rescaling

B9 [Solved]: Prediction-Powered Inference with a Few Human Labels — Task

- **Question:** can a few hundred human labels plus a biased model give a valid and narrower interval for the mean tone of the headlines?
- **Data**
 - ▶ Target θ = mean human net tone $(-1, 0, 1)$ of the 2 388 Twitter validation headlines, which act as the population; f = net score of FinBERT or of Qwen2.5-7B, fixed
 - ▶ Design: n labelled headlines (with Y) and $N - n$ unlabelled ones (only f), drawn independently with replacement
- **Tasks**
 - (a) For one draw with $n = 200$, compute the labels-only estimate $\bar{Y}_n \pm 1.96 s_Y / \sqrt{n}$.
 - (b) Compute PPI [Angelopoulos et al. \(2023\)](#): $\hat{\theta} = \frac{1}{N-n} \sum_{\text{unlab.}} f - \frac{1}{n} \sum_{\text{lab.}} (f - Y)$, $\text{SE} = \sqrt{\hat{V}(f)/(N-n) + \hat{V}(f - Y)/n}$, for both models.
 - (c) For $n = 100, 200, 400, 800$, repeat 2 000 times and record the coverage of θ and the mean CI width.
- **Report:** the one-draw estimates and a coverage and width chart
- **Interpretation:** why is the plain model mean $\frac{1}{N} \sum f$ not an estimate of θ ?

B9 [Solved]: Solution



- True $\theta = 0.054$; plain model means FinBERT -0.014 (wrong sign), Qwen $+0.101$: the bias does not shrink as N grows
- (a) Labels only: $0.010 \pm 1.96 \times 0.042$; (b) PPI-FinBERT: rectifier -0.064 , $\hat{\theta} = 0.054 \pm 1.96 \times 0.036$; PPI-Qwen: rectifier -0.012 , $\hat{\theta} = 0.123 \pm 1.96 \times 0.039$
- (c) $n = 100$: coverage 94.7%, 94.3%, 95.4%, width 0.229, 0.204, 0.203 (labels, PPI-FinBERT, PPI-Qwen); gain for large N : 24% and 27%; at $n = 800$ PPI is wider (0.087 vs 0.081)
- Interpretation: the model mean converges to the wrong number; the labelled sample corrects it, and PPI is narrower only when $\text{Var}(f - Y) < \text{Var}(Y)$ and many texts are unlabelled

B10 [Proposed]: Wild Cluster Bootstrap with 16 Stocks — Task

- **Model:** B5 [Solved]
- **Question:** with only 16 stock clusters, which of the nine slopes of B5 remain significant?
- **Data**
 - ▶ FNSPID panel: 18 282 stock-days with news, 16 US stocks, 27.10.2009–18.12.2023; $z_{i,d}$: daily net score, standardised over the panel; r^e : excess return over SPY (%), from data/market
 - ▶ The nine regressions of B5 (three scores $\times h = 0, 1, 2$), SE clustered by stock ($G = 16$)
- **Tasks**
 - (a) Compute the p -value of each slope with t_{15} critical values.
 - (b) Run the wild cluster bootstrap with the null imposed [Cameron, Gelbach & Miller \(2008\)](#): $y_i^* = \bar{y} + w_{g(i)}(y_i - \bar{y})$, Rademacher $w_g = \pm 1$, 9 999 draws.
 - (c) Compute t with two-way clustering, $\hat{V} = \hat{V}_{\text{day}} + \hat{V}_{\text{stock}} - \hat{V}_{\text{day} \cap \text{stock}}$.
 - (d) Adjust the nine bootstrap p -values with Holm at 5%.
- **Report:** a table with t , the three p -values and the two-way t
- **Interpretation:** does Qwen at $h = 2$ survive?

B11 [Proposed]: Specification Curve for the Sentiment Strategy — Task

- ▣ **Model:** B7 [Solved] (a randomisation test)
- ▣ **Question:** over all reasonable variants of the strategy, is there more evidence of predictability than chance would give?
- ▣ **Data**
 - ▶ Strategy of Lecture 15: at the close of $d + 1$ take $\text{sign}(s_{i,d})$ in every stock with news on day d , hedged with SPY, closed one day later; four trades on each day with positions; 3 568 trading days
 - ▶ Grid of Lecture 15: three scores $\times$ thresholds $|s| > b$, $b \in \{0, 0.25, 0.5\}$, $\times$ return day $d + 2, \dots, d + 5$: 36 strategies
- ▣ **Tasks**
 - (a) Compute the mean daily return and t_{NW} of each strategy and plot the sorted t -statistics [Simonsohn, Simmons & Nelson \(2020\)](#).
 - (b) Simulate the joint null: flip the signs of the positions in blocks of 10 signal days (dependent wild bootstrap [Shao \(2010\)](#)), the same flips for all 36, 2 000 times.
 - (c) Compute the joint p -values of the number of $t > 1.96$ and of the median t .
- ▣ **Report:** the sorted curve and the two joint p -values
- ▣ **Interpretation:** what would a "best specification" report hide?

A8 [Proposed]: The Qwen Strategy after Costs — Task

- ▣ **Model:** A7 [Solved]
- ▣ **Question:** the Qwen2.5-7B strategy has a statistically positive gross mean: is it tradable?
- ▣ **Inputs**
 - ▶ Strategy of Lecture 15: at the close of $d + 1$ take $\text{sign}(s_{i,d})$ in every stock with news on day d , hedged with SPY, closed one day later; four trades on each day with positions; 3 568 trading days
 - ▶ Qwen2.5-7B signal: $\hat{\mu} = 5.64$ bp, $t_{NW} = 2.26$ ($L = 8$), $a = 0.950$, gross Sharpe ratio 0.63
- ▣ **Tasks**
 - (a) Compute the break-even cost c^* .
 - (b) Compute $\text{SE}(c^*)$ by the delta method and the 95% CI.
 - (c) Recover σ and compute the net Sharpe ratio at 5 bp.
- ▣ **Report:** c^* , SE, CI, net Sharpe ratio
- ▣ **Interpretation:** does a significant gross mean imply a profit after costs?

B6 [Proposed]: Event Study with the Loughran–McDonald Tone — Task

- **Model:** B5 [Solved]
- **Question:** around negative and positive news days, when does the price move, and does the dictionary separate the two groups as well as FinBERT?
- **Data**
 - ▶ FNSPID panel: 18 282 stock-days with news, 16 US stocks, 27.10.2009–18.12.2023; $z_{i,d}$: daily net score, standardised over the panel; r^e : excess return over SPY (%), from data/market
 - ▶ The daily LM tone is exactly 0 on 48% of stock-days: groups $\text{tone} < 0$ (5 196 stock-days) and $\text{tone} > 0$ (4 288); FinBERT: bottom and top terciles
- **Tasks**
 - (a) For each date, average the excess returns of days $-5, \dots, +5$ over that date's stock-days in each group.
 - (b) Compute the mean over dates of the CAR for $[-5, -1]$, day 0, day 1 and $[+2, +5]$, with t_{NW} .
 - (c) Plot the mean cumulative paths of the four groups.
- **Report:** the table of the four windows and the chart
- **Interpretation:** why is the post-event return positive for both groups [Tetlock \(2007\)](#)?

C1 [Proposed]: Does Headline Sentiment Pay after Costs? — Task

▣ **Model:** A7, B5 [Solved]

▣ **Question:** does a daily strategy on FNSPID headline sentiment earn money after realistic costs?

▣ **Data**

- ▶ Strategy of Lecture 15: at the close of $d + 1$ take $\text{sign}(s_{i,d})$ in every stock with news on day d , hedged with SPY, closed one day later; four trades on each day with positions; 3 568 trading days
- ▶ Positions equally weighted; days without news: return 0; 5 bp per trade; scores FinBERT, Qwen2.5-7B, LM from the saved tables

▣ **Tasks**

- For FinBERT, compute the mean daily gross return, t_{NW} and a moving-block bootstrap 95% CI (blocks of 10 days, 2 000 draws).
- Compute the Sharpe ratio before and after costs, the break-even cost and the Sharpe ratio at 0, 1, 2, 5, 10 bp.
- Repeat for Qwen2.5-7B and LM, and for strong news $|s| > 0.5$ (exploratory); Holm over the three unfiltered strategies, or over all six if you select among them.
- Further step: correct the same-day slope with a random human-labelled sample of stock-days (divide by $\rho(s^*, \hat{s})$, or rectify the OLS moments as in B9).

▣ **Report:** a two-page note with a results table and charts, and a Quantlet folder

▣ **Interpretation:** discuss timing, look-ahead (B7), survivorship of the 16 stocks and crowding [Lopez-Lira & Tang \(2026\)](#)

C3 [Proposed]: Replicating Lopez-Lira and Tang on FNSPID — Task

- ▣ **Model:** C1, B5, B10, B7
- ▣ **Question:** does the published GPT-4 design [Lopez-Lira & Tang \(2026\)](#), run with an open model on the course headlines, find the same reaction and the same drift?
- ▣ **Inputs**
 - ▶ Paper: GPT-4 answers YES, NO or UNKNOWN about each headline; $s = 1, -1, 0$ drives a daily long-short portfolio; overnight drift 0.34% per day, Sharpe ratio 2.97 (Table 1)
 - ▶ Data: FNSPID headlines of the 16 stocks, adjusted closes from data/market; first safe return $d + 2$
 - ▶ Model: Qwen2.5-7B-Instruct [Qwen Team \(2024\)](#), temperature 0, the prompt of Section 5 of the paper word for word
- ▣ **Tasks**
 - (a) Score every headline, parse the first line only, map YES/UNKNOWN/NO to 1/0/ - 1; report the shares and invalid answers.
 - (b) Average by stock-day and form the daily portfolio with the conventions of the next slide, earning the returns of day $d + 2$.
 - (c) Report the mean daily return in bp, t_{NW} , the hit rate, the Sharpe ratio, and the shares of days with one leg and with no position.
 - (d) Estimate Eq. (1) at headline level for $h = 0, 2$, with stock and date fixed effects, SE clustered by date and stock and the wild cluster bootstrap p (B10).
 - (e) Recompute the Sharpe ratio net of 5, 10, 20 bp per round trip (Figure 4) and compare with C1 on the same stock-days.
- ▣ **Report:** your numbers next to Tables 1 and 2 of the paper, cumulative returns before and after costs, a Quantlet folder
- ▣ **Interpretation:** which differences from the paper can your data separate, and what does B7 say about the bias?

C3 [Proposed]: Portfolio, Calendar and Cost Conventions

□ Fixed before any run

- ▶ Weights (Table 1): long leg = equal-weighted mean return of the stocks with $s > 0$, short leg = same for $s < 0$, a leg only with at least two stocks; long-short = long minus short (gross exposure 2); if only one leg qualifies, hold it alone (footnote 20)
- ▶ Returns: raw stock returns, as in Table 1; the SPY-hedged version (as in C1) is a course adaptation, reported as a sensitivity: with both legs the hedge cancels, so it matters only on one-leg days
- ▶ Calendar: all trading days, as in C1, with return 0 without a position; mean, t_{NW} and Sharpe ratio over all days; hit rate over days with a position
- ▶ Costs (Figure 4): c bp per round trip on each unit of notional: $2c$ per day with two legs, c with one, plus, in the hedged version, c on the net SPY notional (1 on one-leg days)
- ▶ Regressions (d): raw returns in % per day, as in Table 2; the date fixed effects absorb the market return; the wild cluster bootstrap keeps both fixed effects in the null model

References (1/3)

- ▣ Aigner, D.J. (1973). Regression with a binary independent variable subject to errors of observation. *Journal of Econometrics*, 1(1), 49–59.
- ▣ Angelopoulos, A.N., Bates, S., Fannjiang, C., Jordan, M.I., Zrnic, T. (2023). Prediction-powered inference. *Science*, 382(6671), 669–674.
- ▣ Araci, D. (2019). FinBERT: financial sentiment analysis with pre-trained language models. arXiv:1908.10063.
- ▣ Cameron, A.C., Gelbach, J.B., Miller, D.L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3), 414–427.
- ▣ Dawid, A.P., Skene, A.M. (1979). Maximum likelihood estimation of observer error-rates using the EM algorithm. *Applied Statistics*, 28(1), 20–28.
- ▣ DeLong, E.R., DeLong, D.M., Clarke-Pearson, D.L. (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*, 44(3), 837–845.
- ▣ Dong, Z., Fan, X., Peng, Z. (2024). FNSPID: a comprehensive financial news dataset in time series. *Proceedings of the 30th ACM SIGKDD Conference*, 4918–4927.
- ▣ Fleiss, J.L. (1971). Measuring nominal scale agreement among many raters. *Psychological Bulletin*, 76(5), 378–382.

References (2/3)

- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Glasserman, P., Lin, C. (2023). *Assessing look-ahead bias in stock return predictions generated by GPT sentiment analysis*. arXiv:2309.17322.
- ▣ Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q. (2017). *On calibration of modern neural networks*. ICML; arXiv:1706.04599.
- ▣ Lopez-Lira, A., Tang, Y. (2026). *Can ChatGPT forecast stock price movements? Return predictability and large language models*. *Journal of Financial Economics*, 184, 104335.
- ▣ Loughran, T., McDonald, B. (2011). *When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks*. *Journal of Finance*, 66(1), 35–65.
- ▣ Malo, P., Sinha, A., Korhonen, P., Wallenius, J., Takala, P. (2014). *Good debt or bad debt: detecting semantic orientations in economic texts*. *Journal of the Association for Information Science and Technology*, 65(4), 782–796.
- ▣ McNemar, Q. (1947). *Note on the sampling error of the difference between correlated proportions or percentages*. *Psychometrika*, 12(2), 153–157.
- ▣ Newey, W.K., West, K.D. (1987). *A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix*. *Econometrica*, 55(3), 703–708.

References (3/3)

- ▣ Petersen, M.A. (2009). [Estimating standard errors in finance panel data sets: comparing approaches](#). *Review of Financial Studies*, 22(1), 435–480.
- ▣ Qwen Team (2024). [Qwen2.5 technical report](#). arXiv:2412.15115.
- ▣ Sarkar, S.K., Vafa, K. (2024). [Lookahead bias in pretrained language models](#). SSRN Working Paper 4754678.
- ▣ Shao, X. (2010). [The dependent wild bootstrap](#). *Journal of the American Statistical Association*, 105(489), 218–235.
- ▣ Simonsohn, U., Simmons, J.P., Nelson, L.D. (2020). [Specification curve analysis](#). *Nature Human Behaviour*, 4(11), 1208–1214.
- ▣ Tetlock, P.C. (2007). [Giving content to investor sentiment: the role of media in the stock market](#). *Journal of Finance*, 62(3), 1139–1168.
- ▣ Wang, W., Wei, F., Dong, L., Bao, H., Yang, N., Zhou, M. (2020). [MiniLM: deep self-attention distillation for task-agnostic compression of pre-trained Transformers](#). *NeurIPS*; arXiv:2002.10957.